

Linear Algebra – MAT 2610

Section 3.3 (Cramer's Rule, Volume, Linear Transformations)

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Theorem 7

Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any b in R^n , the unique solution x of $Ax = b$ has entries given by



$$x_i = \frac{\det(A_i(b))}{\det(A)}$$
Two red arrows pointing from the right side of the boxed formula to the text below.

where $A_i(b)$ is the matrix obtained by replacing column i by the vector b



Example

Find the values of s for which this system has a solution, and calculate the solution

$$(8 - s)x_1 - 2x_2 = 5$$

$$4x_1 + (2 - s)x_2 = 3$$

$$Ax = b \quad \begin{bmatrix} 8-s & 2 \\ 4 & 2-s \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\det(A) = (s-6)(s-4)$$

is a sol'n when $s \neq 6$ and $s \neq 4$



$$\begin{aligned}(8-s)x_1 - 2x_2 &= 5 \\ 4x_1 + (2-s)x_2 &= 3\end{aligned}$$

$$A_1(b) = \begin{bmatrix} 5 & -2 \\ 3 & 2-s \end{bmatrix}$$

$$\det(A_1(b)) = (5)(2-s) - (-2)(3) = 10 - 5s + 6 = -5s + 16$$

$$\text{so } x_1 = \frac{-5s+16}{(s-6)(s-4)}$$

$$A_2(b) = \begin{bmatrix} 8-s & 5 \\ 4 & 3 \end{bmatrix}$$

$$\det(A_2(b)) = (8-s)(3) - (5)(4) = 24 - 3s - 20 = -3s + 4$$

$$\text{so } x_2 = \frac{-3s+4}{(s-6)(s-4)}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{-5s+16}{(s-6)(s-4)} \\ \frac{-3s+4}{(s-6)(s-4)} \end{bmatrix}$$



Theorem 8

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Let A be an invertible $n \times n$ matrix. Then

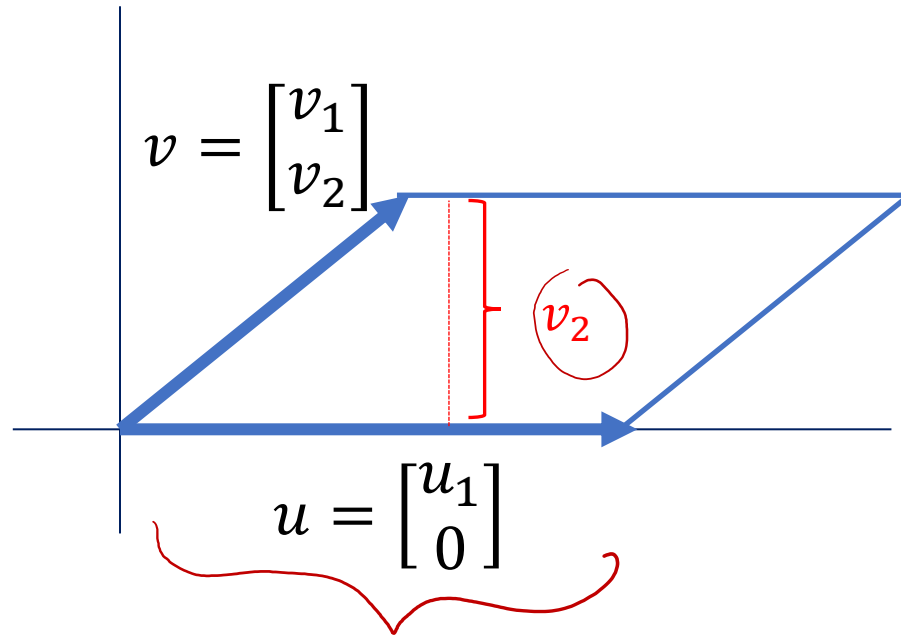
$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

where $\text{adj}(A)$ is the transpose of the matrix of cofactors

$$\text{adj}(A) = \begin{bmatrix} C_{11} & \cdots & C_{n1} \\ \vdots & \ddots & \vdots \\ C_{1n} & \cdots & C_{nn} \end{bmatrix}$$



Geometric Interpretation: Any two non-parallel vectors in R^2 define a parallelogram.



$$\text{Area} = (\text{base})(\text{height}) = \underline{\underline{u_1 v_2}}$$

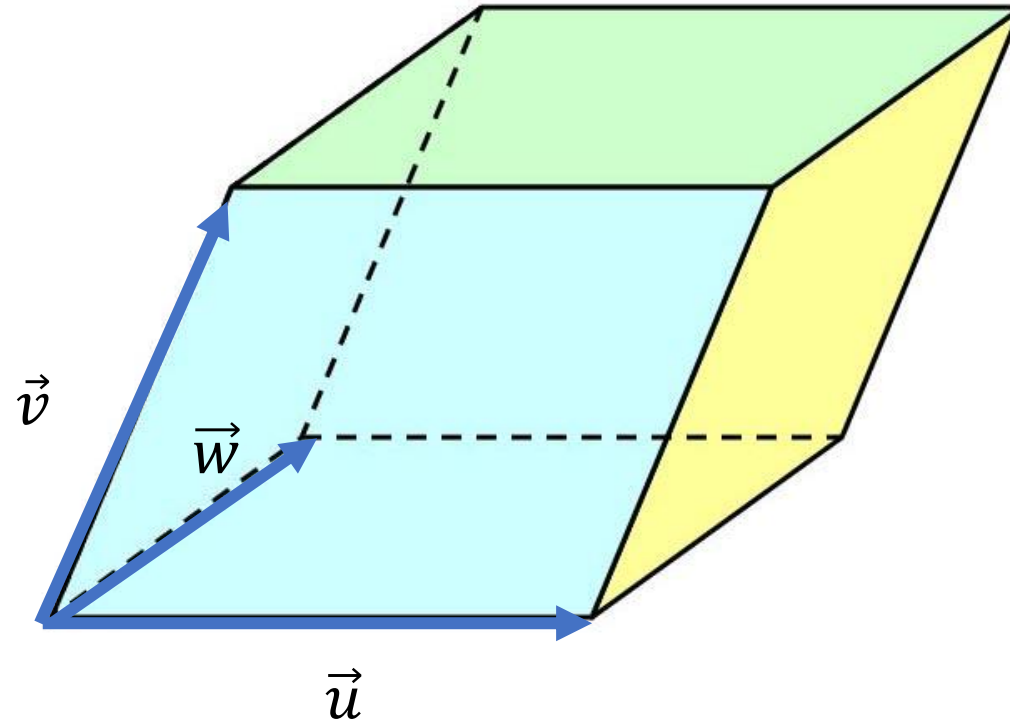
$$\rightarrow \det \begin{bmatrix} u_1 & v_1 \\ 0 & v_2 \end{bmatrix} = \underline{\underline{u_1 v_2 - 0 * v_1}} = \underline{\underline{u_1 v_2}} = \underline{\underline{\text{Area of parallelogram}}}$$

$$\det \begin{bmatrix} v_1 & u_1 \\ v_2 & 0 \end{bmatrix} = v_1 * 0 - u_1 v_2 = -u_1 v_2 = \underline{\underline{-\text{Area of parallelogram}}}$$

$\uparrow$ $\uparrow$



Geometric Interpretation: Any three linearly independent vectors in R^3 define a parallelepiped.



$$| \det[\vec{u} \quad \vec{v} \quad \vec{w}] | = \text{Volume of parallelepiped}$$



Theorem 9

Let A be a 2×2 matrix. Then the area of the parallelogram determined by the columns of A is $|\det(A)|$

Let A be a 3×3 matrix. Then the area of the parallelepiped determined by the columns of A is $|\det(A)|$



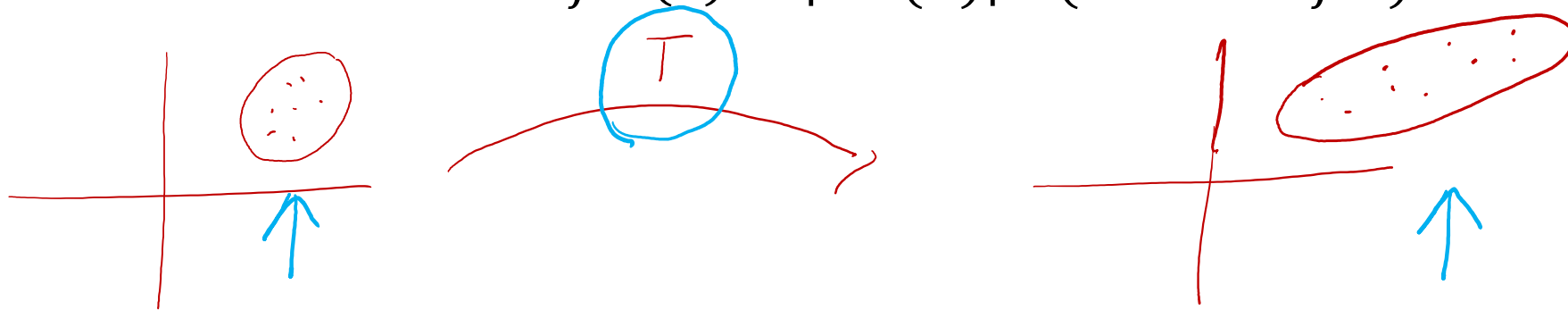
Theorem 10

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A .
If S is a parallelogram in $\mathbb{R}^2$ then

$$\text{area of } T(S) = |\det(A)| * (\text{area of } S)$$

If T is determined by a 3×3 matrix A and S is a parallelepiped in $\mathbb{R}^3$, then

$$\text{volume of } T(S) = |\det(A)| * (\text{volume of } S)$$



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